



Physiologically based pharmacokinetic modeling of polychlorinated biphenyl-153: cross-species extrapolation and human exposure simulation

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Abstract

Polychlorinated biphenyl-153 (PCB-153) is a representative organic pollutant that can accumulate in the human body, particularly in adipose tissue, for long periods due to its environmental persistence and high lipid solubility. Existing risk assessments primarily rely on simple extrapolations based on blood concentrations, and a quantitative approach that considers internal exposure at the tissue level is lacking. The main objective of this study was to establish a human physiologically based pharmacokinetic (PBPK) model for PCB-153 and to quantitatively predict tissue accumulation in the human body based on this model, thereby suggesting the possibility of expanding its application as a precise and scientific human risk assessment tool in the future. Based on existing mouse experimental data, a PBPK model was constructed and validated, and a whole-body human PBPK model was established through interspecies extrapolation. Monte Carlo simulations were then performed, considering physiological variations in the adult population, to predict concentration–time curves for major human tissues (fat, liver, brain, skin, and plasma) following a single dose (20 mg/kg) and repeated exposures at the reference dose (RfD) level (5.86 ng/kg/day). The simulation results based on the human PBPK model established in this study showed that adipose tissue showed the highest accumulation pattern with an average partition coefficient (K_p) of 795,725 after a single dose (20 mg/kg), and even under repeated exposure conditions (RfD), the average K_p in adipose tissue was found to be 4961, the highest among all tissues. In addition to adipose tissue, PCB-153 was widely distributed in the brain, skin, liver, etc., and the same pattern was observed in terms of steady-state concentrations upon repeated administration. These results strongly suggest the possibility of PCB-153 accumulation in major organ tissues even at relatively low external exposure doses (RfD), and support the need for a conservative and quantitative risk assessment, especially for the possibility of chronic toxicity due to long-term exposure. The human PBPK model established in this study establishes a foundation for numerically quantifying the tissue-specific accumulation characteristics of PCB-153, suggesting an alternative approach that overcomes the limitations of existing, limited risk assessments. This model can serve as a scientific basis for future human exposure assessments and standardization of highly soluble environmental pollutants.

Keywords Polychlorinated biphenyl-153 · Tissue distribution · Physiologically based pharmacokinetic model · Interspecific extrapolation · Reference dose · Risk assessment

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Abbreviations

PCB	Polychlorinated biphenyl
PBPK	Physiologically based pharmacokinetic
RfD	Reference dose
POP	Persistent organic pollutants
$T_{1/2}$	Half-life
ADME	Absorption: distribution: metabolism: and excretion
AUC_{inf}	Area under the curve to infinity

C_{last}	Concentration at the last time point
C_{max}	Maximum concentration
T_{max}	Time to reach C_{max}
MRT	Mean residence time
V_d	Volume of distribution
C_{trough}	Minimum concentration at steady state
VPC	Visual predictive check

Introduction

Polychlorinated biphenyls (PCBs) were widely used for industrial purposes until the mid-twentieth century, and have since been considered representative persistent organic pollutants (POPs) due to their high stability and persistence in the environment (Jones and de Voogt 1999; Sandu et al. 2025). Among these, PCB-153 (2,2',4,4',5,5'-hexachlorobiphenyl) is one of the isomers with the highest residual levels in the environment and in living organisms, and various biological toxic effects have been reported (Grilo et al. 2014; Safe 1994). In particular, because it has a strong lipophilicity and a very long biological half-life ($T_{1/2}$) exceeding several years, it is highly likely to accumulate in the body, raising concerns about long-term health effects after exposure (Ritter et al. 2011; Weijs et al. 2010). PCB-153 has been repeatedly detected in serum and adipose tissue in the general population, and according to the US NHANES survey, the average blood PCB-153 concentration was found to be 14.5–19.8 ng/g lipid (Crinnion 2010). Even in the WHO/UNEP Global Survey, serum concentrations of PCB-153 varied widely, ranging from 10 to 83 ng/g lipid in Europe and North America (McLachlan et al. 2018; Undeman et al. 2018). In South Korea, according to the 5th Environmental Health Basic Survey, the average blood exposure level of PCB-153 for adults was 7.15 ng/g lipid (ME 2024).

PCB-153 can indirectly inhibit or activate the cytochrome P450 system, a liver metabolic enzyme, and these actions are associated with hepatotoxicity, endocrine disruption, developmental toxicity, and reproductive toxicity (Ferrante et al. 2014; Lindell 2012). It has also been reported that it can induce intracellular oxidative stress and inhibit mitochondrial function, thereby inducing mechanisms related to apoptosis or carcinogenesis (Glauert et al. 2008). Recently, animal experiments have shown that PCB-153 can reach the brain and nervous system, and can also cause neurotoxicity and neurobehavioral changes (Klocke and Lein 2020; Seelbach et al. 2010). Despite the elucidation of various toxicity mechanisms, quantitative prediction and interpretation tools for the tissue distribution and in vivo behavior of PCB-153 are extremely limited.

Previous studies on PCB-153 have been limited to environmental concentration monitoring, biological sample analysis, epidemiological correlation analysis, or in vitro and in vivo toxicity assessments. Model-based approaches that can quantitatively explain or predict systemic tissue distribution and exposure levels have been virtually nonexistent. In particular, a physiologically based pharmacokinetic (PBPK) model that can provide a quantitative link between exposure levels and tissue accumulation has not yet been reported for PCB-153. This gap presents a major obstacle to reliable human dose prediction from a risk assessment perspective. Indeed, numerous toxicology and environmental health researchers have raised the urgent need for a preclinical model-based quantitative interpretation system for PCB-153 (Chain 2015; Organization 2010).

In this study, we newly developed a PBPK model for PCB-153. We first established a PBPK model for mice and then performed interspecies extrapolation to humans based on this. In particular, we calibrated the model using existing tissue-specific distribution data for mice, and implemented a PBPK model capable of predicting tissue-specific exposure in the body, thereby providing a scientific basis for future risk assessment of PCB-153. In addition, we used this model to estimate the concentration distribution patterns and PK parameters of PCB-153 in major tissues such as brain, liver, fat, skin, and plasma at the previously reported human reference dose (RfD) level (Ermler and Kortenkamp 2022).

The scientific significance of this study can be emphasized in three key aspects. First, this study developed the first whole-body, tissue-based PBPK model for PCB-153, providing a quantitative framework for predicting bioaccumulation and tissue-specific concentrations. Second, by extrapolating the established mouse model to humans, we demonstrated its applicability as a foundational technology for future quantitative human health risk assessments. Third, to validate the model, we evaluated its PK fit and reliability from various perspectives, including residual analysis, two-fold error analysis, and sensitivity analysis. This represents a significant contribution to toxicological quantification, distinguishing it from conventional qualitative and technical approaches. In this respect, this study is an attempt to provide a scientific framework for quantitatively approaching the accumulation and hazard prediction in major human tissues due to chronic exposure to PCB-153, and has great academic and practical significance in that it develops a precise toxicodynamic analysis tool for setting safety standards for environmental toxicants and predicting their effects on the human body.

Methods

Collecting modeling requirements data for PCB-153

To build a PBPK model for PCB-153, we first systematically collected physicochemical and biochemical property information that could be used as modeling inputs for the compound based on the literature. Because these inputs directly impact the structural validity and predictive accuracy of the PBPK model, a rigorous information selection process utilizing reliable data sources was required. The literature search was conducted using major science-based databases such as PubMed, ScienceDirect, Google Scholar, and ECHA Substance Info, and the search strategy was established centered on the following keywords: “PCB-153,” “Pharmacokinetics,” “ADME (absorption, distribution, metabolism, and excretion)” “Tissue distribution,” “Biochemical parameters,” “Partition coefficient,” “PBPK modeling,” “Absorption,” “Metabolism,” “Clearance,” and “Lipophilicity.” An initial literature screening of over 150 papers was conducted. Review papers that did not provide direct numerical information (data-based parameter values) on the PK behavior or modeling of PCB-153, or tissue-specific distribution/metabolism information, environmental monitoring-focused studies, or papers that only addressed congeners were excluded. Furthermore, papers that comprehensively addressed “PCBs” were excluded if PCB-153 was not explicitly addressed as the primary compound.

To obtain the essential in vivo time-course tissue concentration data for the development and validation of the PBPK model, we searched for reports on mouse tissue distribution experiments on single components of PCB-153 accessible within the literature screening index. Among these, the study by Lee et al. (Lee et al. 2002) provided data on the changes in concentrations in liver, brain, fat, and skin tissues over time following a single oral administration of 20 mg/kg of PCB-153 to non-pregnant adult female mice. This study was utilized as the most crucial experimental report for the model development of this study (which aimed to quantify the level of PCB-153 exposure in the general population).

In this report (Lee et al. 2002), the concentrations of PCB-153 in each tissue (liver, brain, adipose tissue, skin tissue) were measured at time points 1, 6, and 13 days after administration, and at time points 2 and 14 days (liver, brain, adipose tissue) using a gas chromatography-based analysis method, and the concentration unit was expressed as ng/g tissue. The mean value and standard deviation for each tissue at each time point were reported together to enable quantitative comparison in model fitting and validation. In this study, these graph data were digitized using the WebPlotDigitizer program (version 4.7) to secure a dataset, which enabled quantitative comparison with the PBPK

model simulation results. The literature-based physicochemical, biochemical, and experimental tissue concentration datasets obtained in this procedure served as essential foundational information for ensuring structural validity and evaluating parameter adequacy of the PCB-153 PBPK model developed in this study.

Mouse PBPK model development

A PBPK model to quantitatively describe the in vivo tissue distribution and PK behavior of PCB-153 was developed using PK-Sim[®] (version 12, Open Systems Pharmacology Suite), which is a validated, mechanistic PBPK platform widely used in regulatory science and toxicology modeling. PK-Sim[®] employs internally defined differential mass-balance equations to govern substance kinetics across physiological compartments. The modeling target was non-pregnant adult female mice, which closely mirrored the conditions used in the experimental validation (Lee et al. 2002).

The PBPK model follows a whole-body-based architecture, with major tissue compartments including plasma, liver, kidney, brain, fat, skin, lung, GI tract, heart, and muscle. Each tissue compartment is interconnected based on physiological characteristics such as tissue volume, blood flow, fat, and water content, and drug distribution between tissues is implemented according to the tissue-to-plasma partition coefficient (K_p). This architecture was automatically built based on the “Mouse (female, adult)” physiological parameter library provided in the default species template of PK-Sim[®], and the default values were maintained without additional modification.

Drug-specific input parameters consisted of previously collected literature-based physicochemical and biochemical information. These included molecular weight, Log P (n-octanol/water K_p), aqueous solubility, pKa (acid dissociation constant), fraction unbound in plasma, membrane permeability, absorption rate constant, dissolution or degradation rate in the gastrointestinal tract, hepatic clearance, and tissue-plasma K_p . These values were mostly supplemented by literature-based experimental data, predicted values based on similar structural compounds, or structure-based outputs, and were ultimately optimized through model calibration based on fitting between observed and predicted values.

To reflect oral administration (oral gavage) conditions in the model, the “oral administration via gavage” option was applied, and the administered dose was set to 20 mg/kg body weight, the same as the experimental conditions (Lee et al. 2002). Absorption occurs through primary diffusion from the intestine, and the distribution of dissolution and absorption regions in the small and large intestines was automatically determined according to the gastrointestinal

physiology model of PK-Sim[®]. Initial simulations of the model were performed through time-concentration profile generation, and the total simulation period was set to 14 days (336 h) to enable comparison with experimentally reported values (Lee et al. 2002). All simulations were performed with a time step of 0.25 h every 15 min, and the calculated concentration units could be converted to ng/g tissue or µg/L using a concentration unit converter within PK-Sim[®]. After model construction and initial condition setting, no link to MoBi[®] (Open Systems Pharmacology Suite) was used, and complete simulations and subsequent parameter optimization were configured solely with functions within PK-Sim[®].

Model calibration and validation

The developed mouse PBPK model underwent a model parameter calibration process to ensure maximum quantitative agreement with experimental data. This process utilized the built-in parameter identification module within PK-Sim[®], and its primary goal was to optimize the values of key model parameters within realistic ranges so that the

model's key outputs dynamically match literature-based experimental results.

During the calibration process, the parameters selected as correction targets were judged to have high predictive sensitivity and high literature-based uncertainty within the overall model structure. Parameter optimization was performed using a multi-compartment simultaneous fitting method for multiple target tissues (liver, brain, fat, skin) rather than a single-tissue-based fitting method. Model overfitting, which compromises the model's physiological validity by maximizing the predictive fit of only a specific tissue, was intentionally avoided. To this end, the search range for each parameter was initially set to a value based on physiological and biochemical values reported in the literature (Tables 1 and 2) and was adjusted to maintain biological explainability. The goodness-of-fit assessment of the model's calibration results was performed using both qualitative visual comparisons and quantitative statistical indicators. First, for visual evaluation, a visual predictive check (VPC) was performed between the model-predicted concentration–time curves for each tissue and the observed values from the literature. This included model-predicted confidence intervals, prediction intervals, and VPC intervals, and a visual assessment was

Table 1 Summary of previously reported and known in vivo pharmacokinetic (PK) properties of PCB-153

PK main category	Description	References
Absorption	Oral Absorption: PCB-153 is highly lipid-soluble ($\log P \approx 7.2$), resulting in greater than 95% absorption when administered orally ^a Absorption rate: Gastric-intestinal absorption constant $\approx 0.08\text{--}0.42\text{ h}^{-1\text{a,b}}$ Main absorption route: Binds to intestinal lipids and is absorbed via the lymphatic system. Binds to blood VLDL and is distributed throughout tissues Characteristics: PCB-153 can be evaluated to be efficiently and rapidly absorbed during gastroduodenal transit when administered orally and to be distributed systemically	ChemSpider (2025); Lee et al. (2007), (2002); PubChem (2025)
Distribution	Main tissue distribution: PCB-153 accumulates in adipose tissue, with over 70% accumulation ^a K_p : Fat/blood of approximately 300 ^a Plasma protein binding: Over 90% bound to albumin, VLDL, etc. ^a Also distributed to brain, liver, and skin: Liver (hepatic distribution $K_p \approx 10$) ^a , brain ($K_p \approx 2.5$) ^a $T_{1/2}$: In humans, the $T_{1/2}$ in the fat compartment is 10–20 years ^c	ChemSpider (2025); Lee et al. (2007), (2002); PubChem (2025)
Metabolism	Main metabolic organ: Liver Metabolic mechanism: Slow hydroxylation by P450 enzymes such as CYP2B, CYP3A, and CYP2E1 Metabolic rate constant: Approximately $0.00016\text{ h}^{-1\text{a,b}}$ in humans Metabolites: Hydroxylated PCBs (e.g., 3-OH-PCB-153), glucuronic conjugates Characteristics: Very slow metabolism, structurally resistant, leading to long-term accumulation in the body	ChemSpider (2025); Lee et al. (2007), (2002); PubChem (2025)
Excretion	Main excretion route: Bile (liver → intestine → feces) is the primary excretion route, with only small amounts of plasma-soluble metabolites excreted via the kidneys (urine) ^a Specificity of excretion during lactation: More than 40% of PCB-153 in nursing mothers is transferred to the infant through breast milk ^a Enterohepatic circulation: After excretion via the bile, a portion is reabsorbed in the small and large intestine, significantly extending the $T_{1/2}$ ^a Excretion ratio: > 80% bile/feces, < 10% kidneys (urine), < 5% breast milk/sebum, etc. ^a	ChemSpider (2025); Lee et al. (2007), (2002); PubChem (2025)

VLDL Very low-density lipoprotein, K_p Partition coefficient, $T_{1/2}$ Half-life

^a This is confirmed to be derived based on mouse PK experimental data

^b This is confirmed by estimates obtained through modeling

^c Rough estimate

Table 2 Summary of input parameters used in the establishment of the initial physiologically based pharmacokinetic (PBPK) model for PCB-153

Category	Parameters	Units	Input information	References
Physicochemical	Molecular weight	g/mol	360.9	ChemSpider (2025); PubChem (2025)
	Chemical formula	NA	C ₁₂ H ₄ Cl ₆	ChemSpider (2025); PubChem (2025)
	Lipophilicity (Log P; XLogP3 ^a)	NA	7.2	PubChem (2025)
	Solubility at pH 7.0 (at pure water)	μg/L	0.91	Ahmad et al. (2019)
	pKa	NA	NA ^b	ChemSpider (2025); PubChem (2025)
	Hydrogen bond donor count	NA	0	ChemSpider (2025); PubChem (2025)
	Hydrogen bond acceptor count	NA	0	ChemSpider (2025); PubChem (2025)
	Binding protein ^c	NA	Albumin; Lipoproteins	PubChem (2025); Rodriguez et al. (2016)
Absorption	Fraction unbound (F _u)	%	5.0 ^d	PubChem (2025); Rodriguez et al. (2016)
	Specific intestinal permeability (transcellular)	cm/min	5.76	Calculated by PK-Sim [®]
	Parameter m for correlation of intestinal permeability (transcellular)	NA	1.11	Calculated by PK-Sim [®]
	Parameter b for correlation of intestinal permeability (transcellular)	NA	0.00	Calculated by PK-Sim [®]
Distribution	Specific intestinal permeability (paracellular)	cm/min	0.01	Calculated by PK-Sim [®]
	Aqueous diffusion coefficient of drug	cm ² /min	3.07 × 10 ⁻⁴	Calculated by PK-Sim [®]
	Fraction recycled to plasma (from tissues)	NA	1.00	Calculated by PK-Sim [®]
Elimination	Partition rate (between plasma and blood cells)	cm/min	173.96	Calculated by PK-Sim [®]
	Plasma clearance (liver)	L/h/kg	1.45 × 10 ⁻⁵	Calculated by PK-Sim [®]
	Specific clearance (liver)	l/min	1.08 × 10 ⁻⁴	Calculated by PK-Sim [®]
	Specific clearance (endosome)	l/min	0.21	Calculated by PK-Sim [®]
	Plasma clearance (biliary)	L/h/kg	1.31 × 10 ⁻⁵	Calculated by PK-Sim [®]
Formulation	Specific clearance (biliary)	l/min	9.74 × 10 ⁻⁵	Calculated by PK-Sim [®]
	Type	NA	Dissolved	Assumed ^e

NA Not applicable, pKa Acid dissociation constant

^a It implied that the values were derived through prediction and simulation via software

^b Structure-based deterministic values

^c PCB-153 is reported to interact with many proteins in the body, but its primary partners are lipoproteins (~ 15%), followed by albumin (~ 85%)

^d Although specific values have not been reported, it was applied based on estimates that it would be over 90% (protein binding) considering the characteristics of similar structural materials

^e Considering the high solubility and excellent absorption properties in the body, which are common among exposure media of PCB-153

made of how well the observed values overlapped within these intervals. The generated ranges were based on Monte Carlo simulations, which, unlike population protocol-based simulations, means that the predicted results reflect only the variability of the parameters themselves, rather than the physiological diversity between hypothetical populations.

Quantitative validation was conducted from various perspectives using various statistical indicators. First, linear regression analysis was performed on the relationship between observed and predicted values to calculate the R² value and regression coefficient (slope), and systematic overestimation or underestimation of the predicted values was identified. Second, residual analysis was performed to confirm the stability of prediction errors over time and to visually confirm whether residuals tended to be biased toward specific time periods. Third, two-fold error range analysis was performed to determine the proportion of predicted values within two times the observed value, which

served as an important comparative indicator for ensuring the reliability of the PBPK model's PK predictions. Finally, a normality test of residuals (Shapiro–Wilk test) was also performed to determine whether the statistical distribution of prediction errors conformed to normality. Through this procedure, we were able to evaluate from various angles whether the model satisfies both physiological/pharmacodynamic validity and mathematical consistency, and model optimization through calibration served as a key step in developing a model that can quantitatively reproducibly describe the time-concentration distribution kinetics in target tissues.

Sensitivity analysis

In this study, we conducted a sensitivity analysis to quantitatively analyze the impact of key model parameters on prediction results. This analysis aims to improve the

model's predictive reliability and identify key factors that should be prioritized for adjustment during future parameter optimization (calibration). The analysis was conducted using the “Sensitivity Analyses” module built into PK-Sim® and targeted all input parameters constituting the whole-body PBPK model. In the sensitivity analysis process, each parameter was independently varied within a $\pm 10\%$ range relative to the reference value, and the sensitivity of the model's predicted results was assessed accordingly. More specifically, the relative impact of each parameter change on exposure levels in target tissues (liver, brain, fat, skin, and plasma) was quantitatively assessed. To this end, simulations were performed for each tissue, and predicted PK parameters such as AUC_{inf} (area under the curve to infinity) and C_{last} (concentration at the last time point) were quantitatively extracted. The analysis results were interpreted based on the magnitude of the influence of individual parameter changes on the predicted tissue-specific concentration–time curve, and the relative influence was quantified by calculating the sensitivity coefficient. The sensitivity coefficient is expressed as follows: $(\Delta Y/Y)/(\Delta P/P)$, where Y is the model output value and P is the input parameter. This coefficient linearizes the sensitivity of the parameter change to the output result, making it interpretable. A larger value indicates a greater influence of the parameter on the model prediction. The model sensitivity analysis process allowed us to identify which drug-specific parameters had the greatest impact on PCB-153 accumulation or clearance in specific tissues, and this information was useful for rationally selecting parameters to prioritize during subsequent model calibration and extrapolation to human models.

Interspecies extrapolation to human PBPK model

In this study, we performed interspecies extrapolation of a PBPK model of PCB-153, developed and optimized in mice, to adult female humans. This extrapolation was implemented using a standard physiological database of human models provided in PK-Sim®. Default values for physiological parameters, such as body weight, cardiac output, tissue-specific blood flow distribution, and volume, were applied to adult females (average body weight 50–60 kg; for each race). During the interspecies extrapolation process, key biochemical parameters, particularly hepatic clearance and biliary clearance, were converted from mouse to human using an allometric scaling approach that accounted for differences in body mass between species. The scaling exponents used were 0.75 for clearance, 1 for volume of distribution (V_d), and -0.25 for rate constant. Parameters related to biodistribution, such as the tissue-to-plasma K_p and the permeability coefficient, are derived through calculation methods based on the material properties of

PCB-153, and it is known that the difference in biochemical pathways depending on the species is relatively small (Dahl et al. 2010), so the calculation methodology applied in the mouse model was applied identically to the human model. However, all of these calculation methods mechanistically predict tissue partitioning and permeability together with the physicochemical properties of the substance (e.g., Log P , pK_a , molecular weight, etc.) by considering the physicochemical components of each tissue within each species, especially the lipid content, water content, and protein content within the tissue. Although the calculation methods are the same, the coefficient values that are actually reflected are species-specific. Furthermore, physicochemical properties of PCB-153, such as lipophilicity, fraction unbound, pK_a , and solubility, are species-independent and inherent to the substance. Therefore, these values, optimized for the mouse model, were applied to the human model as is. This consistent application was a strategy to enhance model comparability and minimize the potential for parameter adjustments during interspecies extrapolation. Consequently, the human model construction performed in this study followed the same compartment framework structurally as the mouse model, but adjusted only the physiological and biochemical parameters to fit human characteristics, which is consistent with the general approach of preclinical-to-clinical predictive modeling (Patel et al. 2025).

Human model simulation and tissue-level exposure prediction

Based on the established human PBPK model, a simulation was performed to predict the tissue distribution and exposure profile of PCB-153 in humans. The simulation was designed to predict the PKs in humans under single-dose and repeat-dose conditions based on the RfD identified in a recent report (Ermler and Kortenkamp 2022), as well as at the same level as the single dose experimentally applied in mice. The simulation was performed using the Population Simulation module in the PK-Sim® software, which was configured to reflect the PK distribution and variability at the population level by generating a virtual population of 100 individuals with various physiological variables (e.g., body weight, hepatic blood flow, plasma protein concentration, etc.) based on a given adult population profile and performing repeated individual simulations on these populations.

To identify sex-specific PK differences in the composition of the generated population, the same simulations were performed in parallel for each of the three groups (a 1:1 mixed group of males and females, all males, and all females). These sex-based population simulations were intended to quantitatively assess the potential differences in

tissue distribution and body persistence of PCB-153 between sexes during long-term exposure. From the predicted results of the simulations, PK indices such as AUC, maximum concentration ($C_{\max}$), clearance, time to reach $C_{\max}$ ($T_{\max}$), $T_{1/2}$, mean residence time (MRT), V_d , and minimum concentration at steady state (C_{trough}) were individually quantitatively extracted for each tissue (brain, liver, fat, skin, plasma, etc.). Based on these, the exposure dose and residence time characteristics in major tissues in the human body were compared and analyzed. These tissue-specific predictions can be utilized as key information for predicting the possibility of tissue accumulation and the risk of chronic toxicity upon repeated exposure. Furthermore, for the repeated-dose simulation, the model was set up based on fixed-interval dosing conditions, in which a fixed dose is administered at fixed intervals, and the model was structured to identify the steady-state reaching point and accumulation ratio for each condition. The overall model simulation and post-processing analysis were performed using standard module-based functions that can be directly extracted from the PK-Sim® simulation results, which is consistent with the methods utilized in existing PBPK studies.

Results

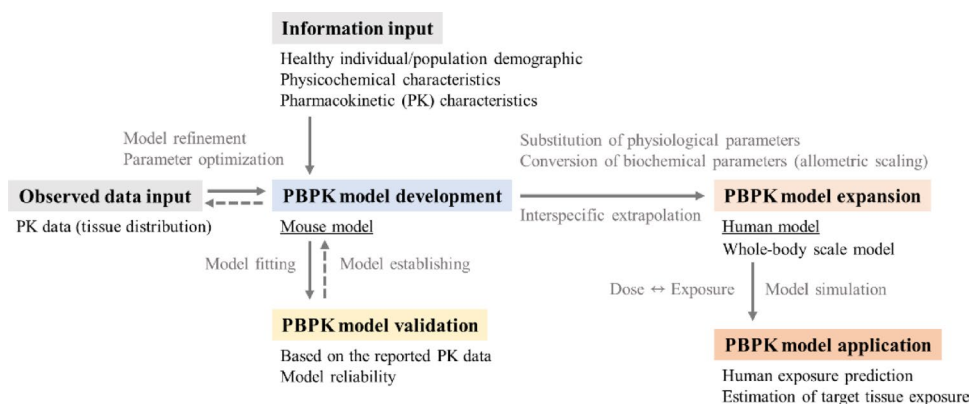
Overall workflow

Figure 1 is a flowchart comprehensively visualizing the development and application of the PBPK model for PCB-153 performed in this study. The overall workflow largely consists of four stages: ① information input and model development, ② model calibration and validation, ③ interspecies extrapolation and human model expansion, and ④ model application, including prediction of tissue-level concentrations within the human body.

First, the starting point of modeling is the information input, which includes the physicochemical properties of PCB-153 (Log P, water solubility, plasma protein binding rate, etc.), physiological parameters (tissue blood flow,

volume, etc.), biochemical parameters (hepatic clearance, etc.), K_p , and cellular permeability, and is mostly composed of default values provided in PK-Sim® and values based on literature reports (Tables 1 and 2). Based on this information, PBPK model development was performed on an initial mouse model, and parameter optimization and model refinement were iteratively performed through comparison with observed data input. This process is schematically represented by the solid and dashed arrows in Fig. 1, and is iteratively adjusted to maximize the agreement between observed and model-predicted values. The next step, PBPK model validation, evaluates how well the developed model quantitatively reproduces the time-concentration curve from independent external experimental data. Model validity is confirmed through model fit and predictive power verification. The validated model then moves on to the PBPK model expansion stage, where it is extended to the human model through interspecific extrapolation. Physiological parameters are replaced with human-specific values, and biochemical parameters are converted through allometric scaling as needed. The K_p and permeability coefficient calculation algorithms utilize the same calculation method to account for species specificity, but species-specific adjustments are made based on the lipid/water composition ratio of each tissue. Finally, in the PBPK model application stage, the established human body model is used to quantitatively analyze the dose-exposure relationship through single and repeated exposure simulations and to predict tissue-level exposure indices (AUC, $C_{\max}$, MRT, clearance, etc.). These prediction results serve as the basis for evaluating tissue-specific accumulation patterns and the potential for toxicity. Figure 1 structurally presents how the PBPK model can be applied as an integrated analysis platform to quantitatively predict human tissue-level cumulative exposure to toxicants through organic linkage between experimental toxicity data and numerical models, going beyond simple pharmacodynamic predictions.

Fig. 1 Schematic diagram of the development, validation, and application processes of the PCB-153 physiologically based pharmacokinetic (PBPK) model performed in this study. The solid lines in the schematic diagram represent the movement flow of each major process, and the dotted lines represent the movement to the next step through the iterative processes



PBPK model construction and input parameters for PCB-153 in mouse

Table 1 comprehensively summarizes the ADME information used as a key basis for the design of the PBPK model structure and initial parameter settings for PCB-153. This table is based on literature-based qualitative and quantitative data, and each item is reflected as a key parameter within the model, playing a central role in setting input values for predicting the material's *in vivo* behavior.

First, in terms of absorption, PCB-153 is a lipophilic substance with a Log P value of approximately 7.2, demonstrating rapid and extensive gastrointestinal absorption through oral administration. This high absorption rate was considered when determining the intestinal permeability coefficient and absorption rate constant within the PBPK model. In terms of distribution, it was emphasized that PCB-153, due to its high solubility, accumulates primarily in adipose tissue, with the lipid layer, rather than plasma, being its primary distribution site. This distribution characteristic was reflected in the determination of the K_p , and concentrations in adipose tissue in humans and rodents have been reported to be hundreds of times higher than in plasma (Lee et al. 2007, 2002). Furthermore, protein binding is very high, with PCB-153 primarily present in plasma bound to lipoproteins or albumin, with a low free concentration. In terms of metabolism, PCB-153 has a relatively resistant structure to metabolism, which is related to the slow oxidative metabolism by CYP450 enzymes due to its high chloride substitution structure. Although PCB-153 is converted to hydroxylated metabolites in the liver, its metabolism is relatively slow. This low intrinsic clearance value necessitates a conservative approach when determining hepatic clearance and $T_{1/2}$. The excretion section emphasizes that PCB-153 is primarily excreted via the bile and feces, with only minimal excretion via the kidneys. In experimental animal studies, 80–90% of the total administered dose is excreted via the hepatic-biliary-intestinal route, with some influence of enterohepatic circulation also reported (Kania-Korwel and Lehmler 2016; Peterson et al. 1976). Thus, Table 1 comprehensively describes the ADME characteristics of PCB-153 in the body, serving as the foundation for establishing model input values and ensuring the physiological validity of modeling scenarios. Furthermore, this ADME information served as a key indicator for quantitatively adjusting tissue distribution ratios, metabolic rates, and tissue clearance within the PBPK model.

Table 2 summarizes the key initial parameter values applied to the human PBPK model of PCB-153 established in this study, which were derived from web-based databases (e.g., PubChem, ChemSpider), existing relevant literature, and the default built-in database of PK-Sim®. First, the

chemical structure containing halogen elements and the Log P value (7.2) reflect the high lipophilicity of PCB-153, suggesting the possibility of accumulation in tissues with high lipid content in the body (e.g., adipose tissue). The plasma protein binding rate is very high at 95%, indicating that the free fraction of PCB-153 in the body is extremely limited, a factor that can significantly affect the rate of tissue diffusion and clearance. As for absorption-related parameters, the transcellular intestinal permeability was set to 5.76 cm/min to reflect the intestinal absorption rate in a specific organ, which is similar to the absorption characteristics of highly soluble compounds suggested in the existing literature. The extent of elimination by hepatic metabolism and biliary excretion was set to low values ($1.08 \times 10^{-4} \text{ min}^{-1}$ and $9.74 \times 10^{-5} \text{ min}^{-1}$), respectively, considering literature-based data on the persistence of PCB-153 in the body, which is an important factor explaining that PCB-153 has a very long biological $T_{1/2}$. Renal excretion was judged to be extremely minimal according to existing literature, and therefore renal elimination was not considered as a major route of elimination in this model. In addition, the tissue K_p values were calculated based on the PK-Sim standard methodology in PK-Sim®, taking into account the lipid and water content of each tissue, and these values served as key inputs for tissue-specific exposure prediction. The dosage form information was applied as a solubility model based on the fact that highly fatty PCB-153 should be mixed with lipid substances to facilitate absorption in the body. All these parameters were configured to quantitatively reflect the *in vivo* kinetic characteristics of PCB-153 and were used as basic data for future tissue-level exposure prediction and repeated-dose simulation.

Model calibration and goodness-of-fit in mouse

Table 3 presents a summary of the final optimized model parameters derived through the model calibration process for developing the PCB-153 PBPK model in mice, and includes details of the parameters that were calibrated based on literature-based initial values to improve the predictive power of the model.

For the K_p , the values were set for major tissues such as brain, liver, adipose, and skin, and it can be confirmed that the calibration results are reflected by comparing them with literature-based values for each tissue. The K_p calculation method for each tissue was ultimately fixed to the same PK-Sim standard method that was initially applied. In addition, cellular permeabilities related to body absorption were optimized from the initially applied PK-Sim standard method to the charge-dependent Schmitt method. This correction is intended to improve the prediction accuracy of the model for tissue distribution and body absorption. In particular, for

Table 3 Summary of optimization of input parameters applied to establish the final physiologically based pharmacokinetic (PBPK) model of PCB-153

Category	Parameters	Units	Input information	Optimized
Physicochemical	Effective molecular weight ^a	g/mol	228.88	Calculated by PK-Sim [®]
	Lipophilicity (Log P)	NA	7.00	From the initial value of 7.2 to the final optimized value of 7.00
	Solubility at pH 7.0	µg/L	0.79	From the initial value of 0.91 to the final optimized value of 0.79
	Fraction unbound (F_u)	%	8.70	From the initial value of 5.0 to the final optimized value of 8.70
Absorption	Cellular permeabilities calculation	NA	Charge dependent Schmitt method	From the initial application PK-Sim standard to the final optimization Charge dependent Schmitt
	Parameter m for correlation of intestinal permeability (transcellular)	NA	1.11	No significant changed ^b
Distribution	Partition coefficients calculation	NA	PK-Sim standard method	No significant changed ^b
	Partition rate (between plasma and blood cells)	cm/min	173.96	No significant changed ^b
Elimination	Plasma clearance (liver)	L/h/kg	1.45×10^{-5}	No significant changed ^b
	Specific clearance (liver)	1/min	1.08×10^{-4}	No significant changed ^b
	Plasma clearance (biliary)	L/h/kg	1.31×10^{-5}	No significant changed ^b
	Specific clearance (biliary)	1/min	9.74×10^{-5}	No significant changed ^b

NA Not applicable

^a It refers to the value according to the removal of halogen elements within the structure

^b This means that there was no significant change between the information applied when establishing the initial model and the information optimized in the final modeling process

the liver and adipose, the initial errors of more than threefold were significantly corrected to within threefold, reflecting the characteristics of PCB-153's rapid absorption and wide distribution among tissues. In addition, parameters related to the physicochemical properties of PCB-153 were also fine-tuned (minor optimization) through calibration. Log P is an important indicator reflecting the characteristics of PCB-153 as a highly soluble substance, and is reported to range from approximately 6.8 to 7.2 in web-based databases (Table 2). Accordingly, the initial value in this model was set to 7.2, and it was ultimately optimized to 7.0, which was adjusted to a reasonable level within the range of theoretical values and literature estimates. This indicates that the model appropriately reflects the distribution of lipid compartment centers. In the case of solubility, the initial value of 0.91 µg/L was reduced by approximately 13% to 0.79 µg/L. This value is still within the 20% variation range. Considering the lack of clear experimentally-based solubility data for PCB-153 and the fact that the initial value itself utilized a prediction based on molecular structure and Log P, this variation is considered a re-estimation within an acceptable range. Therefore, this can also be interpreted as a physiologically valid correction result. For plasma protein binding (fraction unbound, f_u), the initial value was 0.05 (95% binding), and the optimized value was adjusted to 0.087 (91.3% binding). PCB-153 has been reported to exhibit high binding properties to VLDL, albumin, etc. in vivo, and previous studies have suggested that the total binding rate is over 90%

(Tables 1 and 2). Therefore, $f_u = 0.087$ applied in this model can be accepted as a reasonable level that is consistent with the structural characteristics and literature evidence, even considering the uncertainty of the experimentally based values. The key parameters related to distribution and clearance—such as distribution rate, hepatic and biliary clearance—did not differ significantly between the initial values and the values after model optimization, suggesting that the structural fit of the model (especially the clearance part) was high. This means that the initial model already had sufficient explanatory power for the physiological and pharmacological properties, and that during the calibration process, it was able to well explain the observed data with only fine adjustments of parameters rather than the structure itself. In summary, the results in Table 3 demonstrate that the model possesses structural validity and physiological consistency, sufficiently describing the biodistribution and elimination kinetics without excessive parameter tuning across the model. Furthermore, the adjusted physicochemical parameter values are optimized to a level that is sufficiently reliable within the literature-based numerical and experimental uncertainty ranges. Therefore, the PBPK model is considered to possess both biological validity and predictive reliability while maintaining high consistency with actual animal experimental data.

The results for visually evaluating the model's predictive performance are presented in Fig. 2. This figure compares the observed and model-predicted values for the temporal

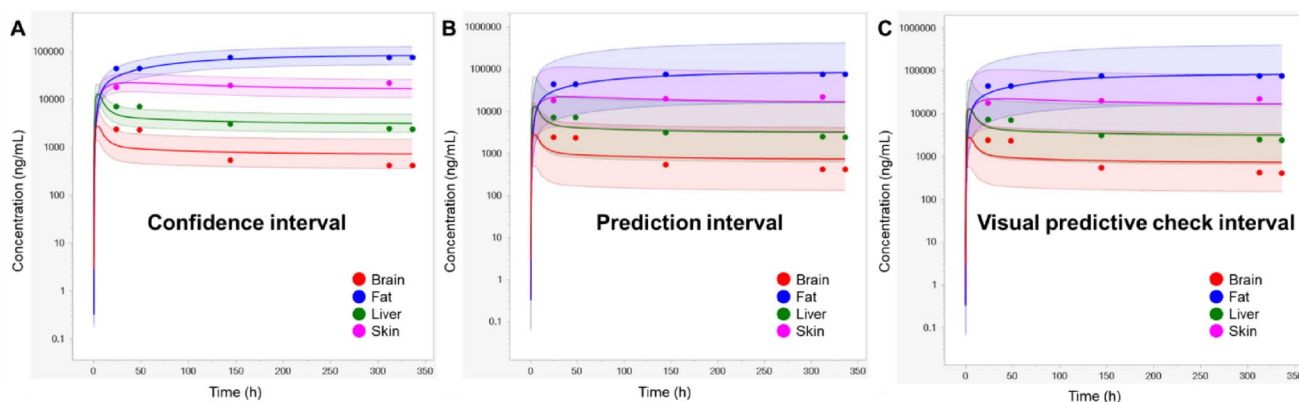
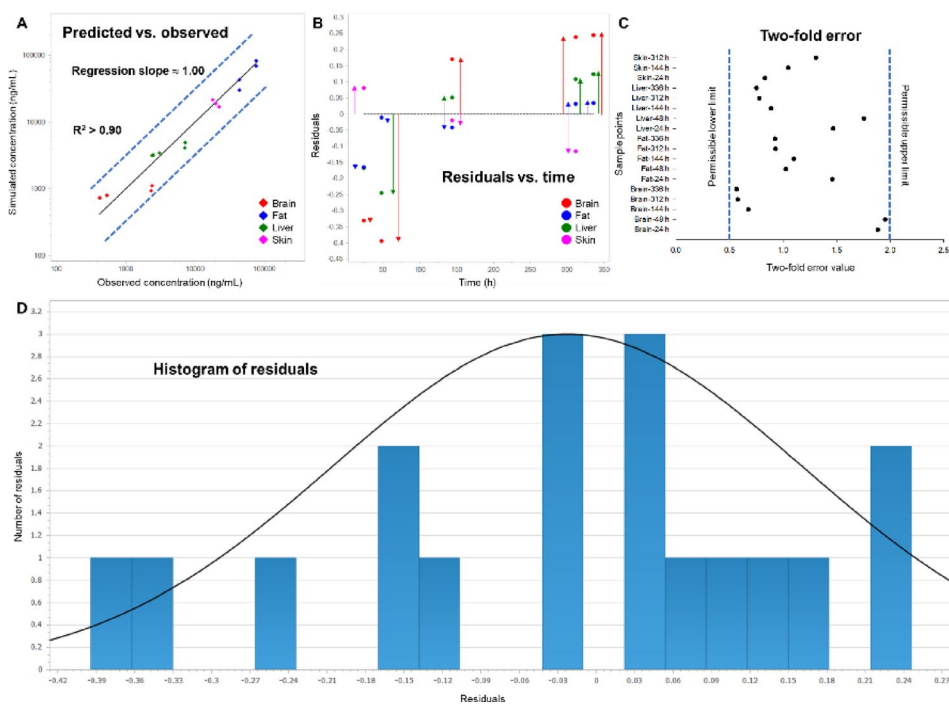


Fig. 2 Visual graphical results of the fit between the observed (after a single 20 mg/kg dose) and predicted values of the mouse physiologically based pharmacokinetic (PBPK) model for PCB-153. Dots in the graph represent observed values, and solid lines represent model predicted values. Shaded areas in the graph represent confidence or

prediction intervals based on the model. **A** 95% confidence intervals compared to observed values; **B** 95% prediction intervals compared to observed values; **C** Visual predictive check intervals compared to observed values

Fig. 3 Validation results of the developed mouse physiologically based pharmacokinetic (PBPK) model for PCB-153. **A** Comparison of agreement between observed values and model predictions (solid line in the graph indicates perfect agreement between predicted and observed values, and the dotted lines indicate the 95% prediction interval); **B** Identification of residual value (difference between model-predicted and measured values) patterns (dotted line in the graph indicates perfect agreement between predicted and observed values, and the solid arrows indicate the direction of the residuals); **C** Identification of two-fold error values for each sample point (dotted lines in the graph indicate acceptable bounds for two-fold error); **D** Histogram of residual value distributions



concentration changes in brain, liver, fat, and skin tissues of PCB-153 after a single oral administration of 20 mg/kg to mice. Figure 2A presents the results compared with 95% confidence intervals, demonstrating that the observed values are well distributed within the predicted curves and confidence intervals. Figure 2B shows the results based on the prediction interval, confirming that most of the observed values fall within the range that includes the uncertainty of the model prediction. The VPC interval in Fig. 2C shows that the prediction range generated through the Monte Carlo-based simulation results, which reflect the uncertainty of the model structure and parameters, sufficiently encompasses the observed values.

A quantitative model fit assessment is presented in detail in Fig. 3. Figure 3A presents the results of a correlation analysis between the observed and predicted values. The regression line slope was approximately 1.00 and the coefficient of determination (R^2) was over 0.90, indicating a very high level of fit. Figure 3B plots the distribution of residuals over time, confirming that there was no biased prediction tendency in specific time periods or organizations, with most residuals falling within the ± 0.3 range. Figure 3C presents the results of a two-fold error analysis for each organization and observation point over time. All data fell within the acceptable $\pm$ twofold error range, suggesting that the model's predictive accuracy was reliable. Finally, Fig. 3D presents a

histogram of the residuals and a normal distribution curve, demonstrating that the residuals were symmetrical around the mean and met the assumption of a normal distribution. This supports the absence of systematic bias in the model prediction error and the overall good representation of the predicted values of the observed values. Overall, these results demonstrate that the mouse-based PCB-153 PBPK model developed in this study was sufficiently calibrated and validated to quantitatively and appropriately reflect the pharmacodynamic behavior in various tissues, including the brain, liver, skin, and fat, and suggest that it can be utilized as a reliable prediction platform for extrapolation to subsequent human models and exposure assessment applications.

Sensitivity analysis of key parameters

Figure 4 visually presents the results of the sensitivity analysis of the main parameters of the PBPK model for PCB-153 constructed in this study, and quantitatively evaluates the factors affecting the variability of the predicted values for each target tissue. The results of the sensitivity analysis

were derived based on the sensitivity coefficient for changes in AUC_{inf} or C_{last} , which serve as prediction indices for each tissue. Only parameters with an absolute value of the sensitivity coefficient of 0.2 or higher among all parameters were selected and presented (Fig. 4). This is to conduct an analysis focusing only on variables that have a substantial impact on the model prediction results.

First, the most sensitive parameter in Fig. 4A (AUC_{inf} in brain) was ‘PCB-153: Lipophilicity (Log P)’, with the highest sensitivity coefficient of -3.7 . This suggests that the high lipid solubility of PCB-153 has a key impact on its accumulation and distribution in brain tissue, and reaffirms that lipid solubility of drugs is a major determinant, especially in tissues that must pass the blood–brain barrier, such as the central nervous system. In addition, ‘Fat: Volume fraction (lipid)’, ‘Brain: Volume fraction (lipid)’, ‘PCB-153: Effective molecular weight’, ‘ K_p (plasma–interstitial)’, ‘Gastrointestinal transit time’, and ‘Jejunum surface area enhancement factor’ all showed $|sensitivity| > 0.2$, confirming that these physiological and physicochemical variables significantly affect the distribution in the brain. Similarly, in

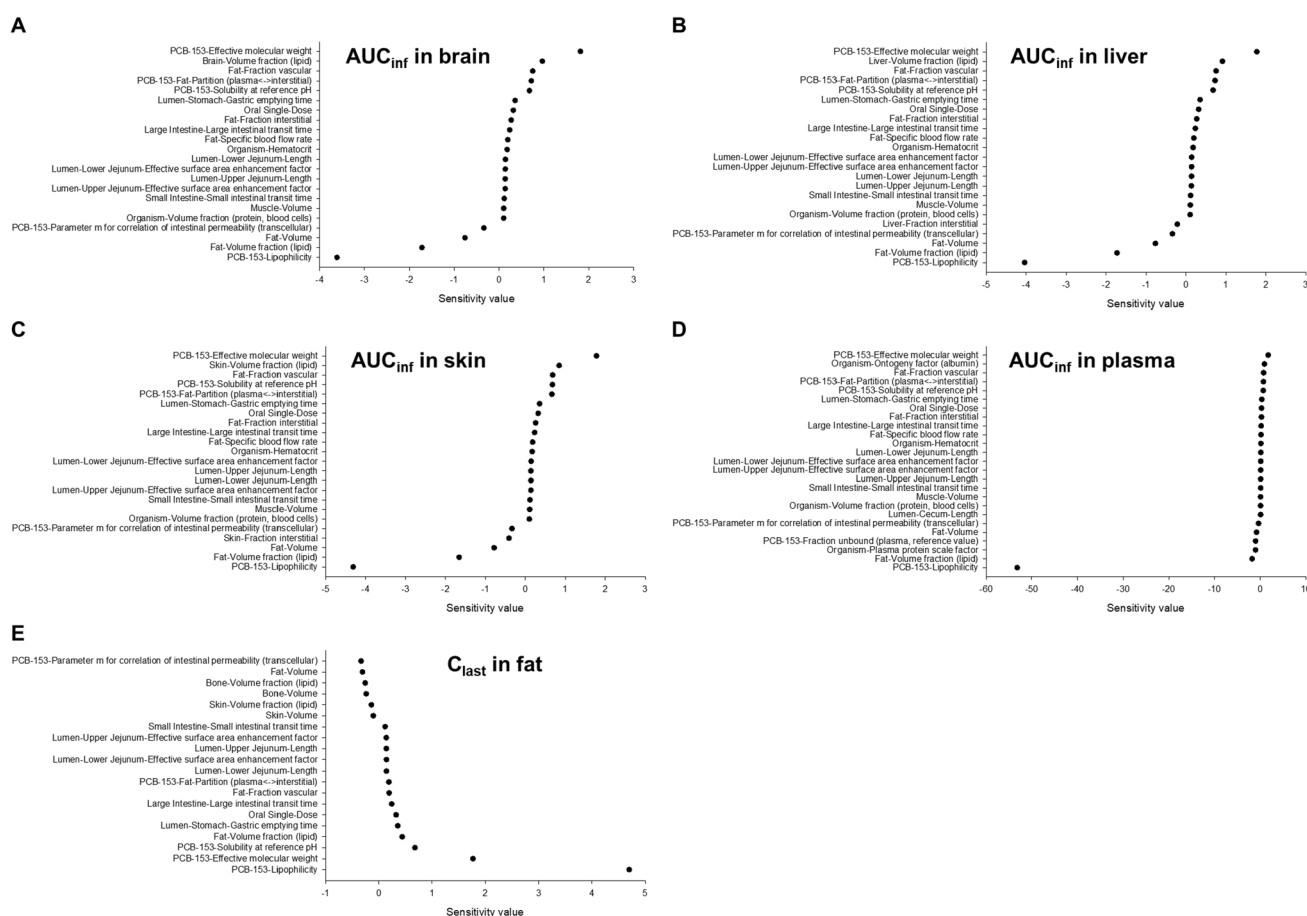


Fig. 4 Sensitivity analysis results for the physiologically based pharmacokinetic (PBPK) model of PCB-153 established in this study. **A** AUC_{inf} in brain; **B** AUC_{inf} in liver; **C** AUC_{inf} in skin; **D** AUC_{inf} in

plasma; **E** C_{last} in fat. AUC_{inf} Area under the profile curve from 0 to infinity after PCB-153 exposure, C_{last} Concentration value at the last observation time after PCB-153 exposure

the analysis results of Fig. 4B (AUC_{inf} in liver), ‘PCB-153: Lipophilicity (Log P)’ showed the greatest influence, and other distribution-related parameters such as ‘Volume fraction (lipid)’ and ‘PCB-153: K_p (plasma-interstitial)’ of liver tissue showed sensitivity coefficients with absolute values ranging from 0.2 to 2. Since the liver is a high-blood flow tissue with a large lipid accumulation capacity, it is interpreted that the sensitivity of the K_p and tissue fat distribution-related factors was relatively greater in predicting the liver distribution of highly soluble environmental residues such as PCB-153. In Fig. 4C (AUC_{inf} in skin), in addition to the lipid-soluble parameter (Log P), ‘Skin: Volume fraction (lipid)’ and ‘Skin: Fraction interstitial’ were derived with sensitivity coefficients of absolute values of 0.2–1, and these variables had a major influence on explaining the accumulation of PCB-153 in the skin. Since the skin has relatively low metabolic activity compared to other tissues and has a structure with a developed lipid layer that can function as a storage of substances, it is interpreted that lipid-soluble and skin-specific physiological factors are ranked high in the sensitivity items. In Fig. 4D (AUC_{inf} in plasma), ‘PCB-153: Lipophilicity (Log P)’ showed the highest sensitivity with a sensitivity coefficient of approximately –53. This result clearly suggests that the degree of lipid solubility is a key determining factor in the systemic exposure of PCB-153. This interpretation is consistent with the pharmacodynamic principle that highly lipid-soluble drugs extensively penetrate, distribute, and accumulate from the blood to adipose tissues. In addition, ‘PCB-153: Effective molecular weight’, ‘Albumin: Ontogeny factor’, and ‘Fat fraction vascular’ also had an absolute value of sensitivity coefficient of 0.2 or higher, affecting the prediction of blood concentration. Finally, in Fig. 4E (C_{last} in fat), various parameters such as ‘Log P’, ‘Fat: Volume fraction (lipid)’, ‘Bone: Volume’, and ‘PCB-153: Solubility’ showed sensitivity coefficients in the range of 0.2–5, and in particular, ‘Log P’ was evaluated as the variable with the greatest influence on predicting the final concentration in fat tissue, with a value of approximately +4.8. This is closely related to the long-term accumulation characteristics of PCB-153, a highly soluble environmental pollutant, in fat tissue, and can be considered a key parameter for model prediction.

Overall, the sensitivity analysis results commonly confirmed that Log P (lipophilicity) showed the highest sensitivity for most tissue prediction values (AUC_{inf} and C_{last}), which served as evidence that the biodistribution dynamics of high-lipophilic chemicals are critically influenced by lipophilicity. In addition, variables related to gastrointestinal absorption (surface area of the jejunum and ileum, transit time, solubility), tissue volume and blood flow, protein-binding factors, and total blood component composition (hematocrit) were also found to have a certain level

of influence on the prediction sensitivity. These analytical results not only predict the in vivo behavior of PCB-153, but also provide guidance on which parameters should be focused on initialization and optimization when building predictive models for similar highly soluble environmental residues. In particular, they can serve as important criteria for prioritizing experimentally derived parameters to enhance model reliability.

Human PBPK model via interspecies extrapolation

In this study, we performed interspecies extrapolation to humans based on the developed mouse-based PBPK model, and constructed an extended human PBPK model based on human physiological parameters and body weight (average adult body weight 60–70 kg; for each race). Then, based on the model, we simulated the temporal concentration patterns of PCB-153 in tissues of a population (50% male and 50% female, $n=100$) following a single oral administration of 20 mg/kg of PCB-153. Figure 5A shows the concentration–time curve in target tissues using this human PBPK model. The simulation results showed that the concentration of PCB-153 in all tissues increased rapidly immediately after administration and then gradually stabilized. In particular, in adipose tissue, a tissue with high lipid content, the initial peak concentration (C_{max}) (within 24 h) was approximately 14.21 $\mu\text{g/mL}$ on average, and the high concentration was maintained thereafter, increasing to 27.25 $\mu\text{g/mL}$. Liver and brain tissues also showed relatively high accumulation characteristics, with an average C_{max} of approximately 39.69 $\mu\text{g/mL}$ and 14.90 $\mu\text{g/mL}$, respectively. Skin was approximately 14.98 $\mu\text{g/mL}$, showing an intermediate concentration distribution compared to other tissues. On the other hand, the concentration in plasma was very low compared to other tissues, and the maximum concentration remained at approximately 0.31 ng/mL. This result is interpreted as related to the high lipophilicity (Log P=7) and high protein binding rate (91.3%) of PCB-153, which causes it to preferentially distribute to lipid tissues. In addition, this result supports that PCB-153 is a representative lipophilic compound that is evenly distributed in most tissues while showing high accumulation in lipid storage tissues. In this simulation, a Monte Carlo-based population model was applied to account for inter-individual variation, and the 2.5th and 97.5th percentiles of concentrations for each tissue are indicated by dotted lines. For adipose tissue, inter-individual variation was relatively large, which is believed to be due to differences in body fat mass and variations in related parameters within the population.

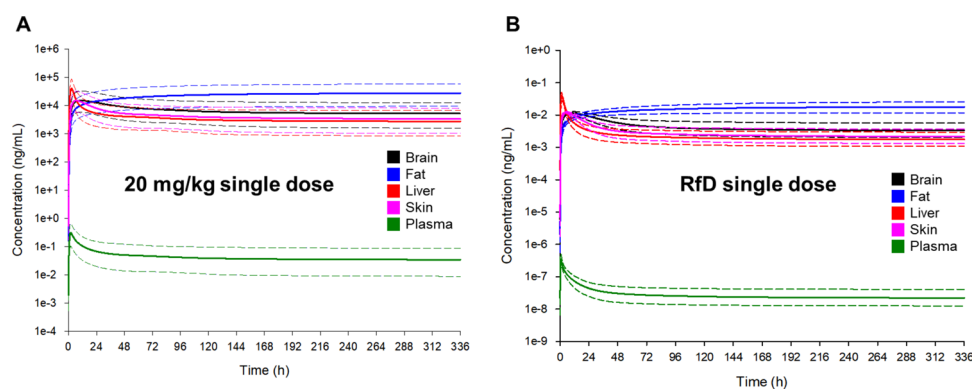


Fig. 5 Results of simulation of the population ($n=100$, comprised of 50% male and 50% female) pharmacokinetic (PK) profiles among target tissues following a single oral administration of 20 mg/kg (A) or reference dose (RfD, B) of PCB-153 using the human physiologically based pharmacokinetic (PBPK) model of PCB-153 established

Simulation of human tissue exposure at RfD

This paper presents various simulation results utilizing a human PBPK model developed to quantitatively understand how external exposure to PCB-153 at the RfD level is distributed and maintained in major tissues within the human body. This study applied an RfD value of $0.0058 \mu\text{g/kg/day}$, proposed through a systematic review of the existing literature, based on recent research findings specifically focusing on semen quality degradation as a key health impact indicator.

First, Fig. 5B visualizes the changes in PCB-153 concentrations in different tissues up to 336 h after a single oral dose of RfD in a mixed-gender group ($n=100$, 50 males and 50 females). All tissues showed a rapid initial absorption followed by a relatively gradual decline, with the highest level of accumulation and a tendency to persist, particularly in adipose tissue. The mean predicted curve (solid line) and the inter-individual variation (dotted line) within the 2.5–97.5% range clearly show the characteristics of the concentration distribution at the group level. Similar concentration profiles were observed in the liver, brain, and skin, but the lowest concentration level was consistently maintained in plasma.

Figure 6 compares the predicted concentration distributions separately in female (A) and male (B) groups under the same RfD single-dose conditions. Each gender group consisted of $n=100$, and a population module was used that applied an average body weight range of 50–60 kg for females and 60–80 kg for males. The simulation results showed that both groups showed the same overall pattern of high concentration accumulation in adipose tissue and low concentration maintenance in plasma. The box plot comparison results in Fig. 6C also suggested that the difference in concentration distribution between the two groups was

not statistically significant. In particular, when compared with weight-normalized values, the difference was further reduced, confirming that the difference in the body distribution of PCB-153 by gender was not significant.

Meanwhile, since continuous exposure to PCB-153 is more common than single exposure in actual environmental exposure, Fig. 7 presents the results of a multiple-dose simulation performed according to a scenario of repeated oral administration at 24 h intervals. This was because the recently reported RfD for PCB-153 is presented in daily units and because it is a standard dosage applied to continuous exposures to substances in general (Ermler and Kortenkamp 2022).

Figure 7A shows the predicted concentration profiles in each tissue for up to 100 days of repeated administration, a total of 2400 h, and clearly shows a pattern of reaching a steady-state in all tissues over time. In particular, gradual accumulation with the highest concentration was observed in adipose tissue, while relatively low concentrations were maintained in plasma. Based on these simulation results, Fig. 7B summarizes the distribution of PCB-153 concentrations in each tissue in the form of a box plot at steady-state point in the 2,376–2,400 h period after repeated administration, and among them, adipose tissue showed the highest concentration level on average (up to approximately 2.5 ng/mL or more), whereas plasma showed the lowest concentration (approximately 0.004 pg/mL or less). Finally, Fig. 7C presents the predicted results converted to ng/g lipid units by applying the average value of blood lipid concentration (based on 650 mg/dL) for the purpose of comparing the PCB-153 concentration in plasma with the actual biomonitoring results. As a result, the predicted steady-state average concentration of PCB-153 converted to the standard plasma lipid unit of a normal adult was approximately 0.0003 ng/g lipid, which was much lower than the actual reported range of blood PCB-153 concentrations in

Fig. 6 Comparison of the pharmacokinetic (PK) profile simulation results among target tissues following a single oral administration of PCB-153 at the reference dose (RfD) for female and male populations ($n=100$), respectively, in the human physiologically based pharmacokinetic (PBPK) model of PCB-153 established in this study. **A** Model simulation results for female population; **B** Model simulation results for male population; **C** Boxplot comparison of PCB-153 concentration distributions between female and male populations. The solid lines in the graph represent the group mean values predicted by the model, while the dotted lines represent the values corresponding to 97.5% (upper) and 2.5% (lower) of the groups, respectively

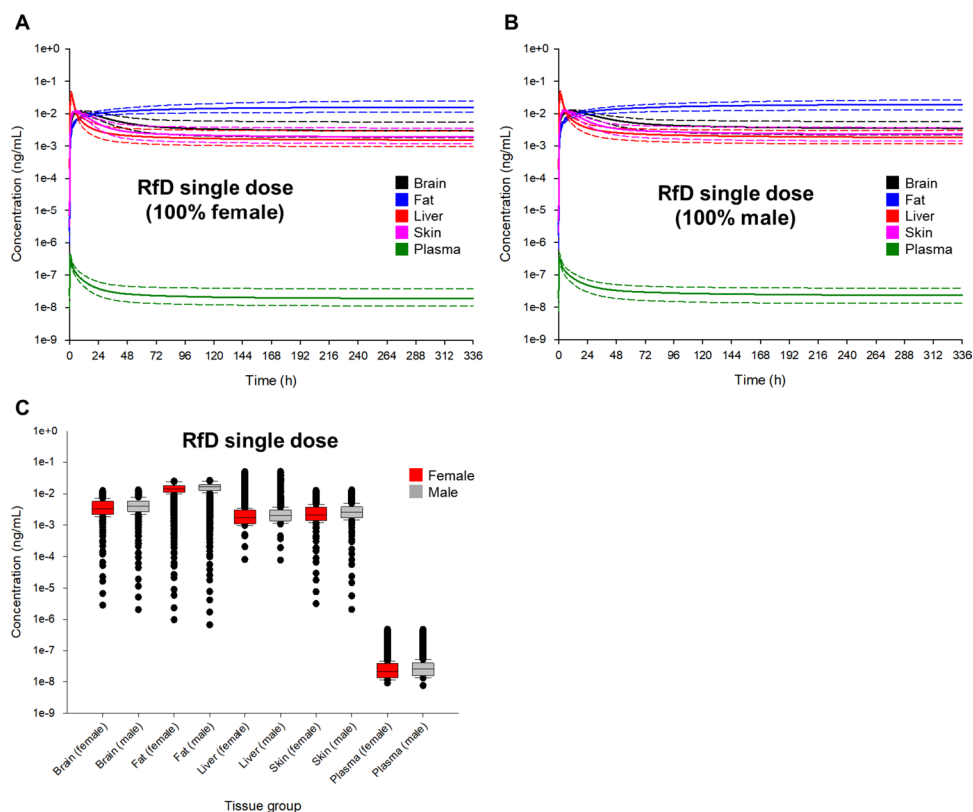
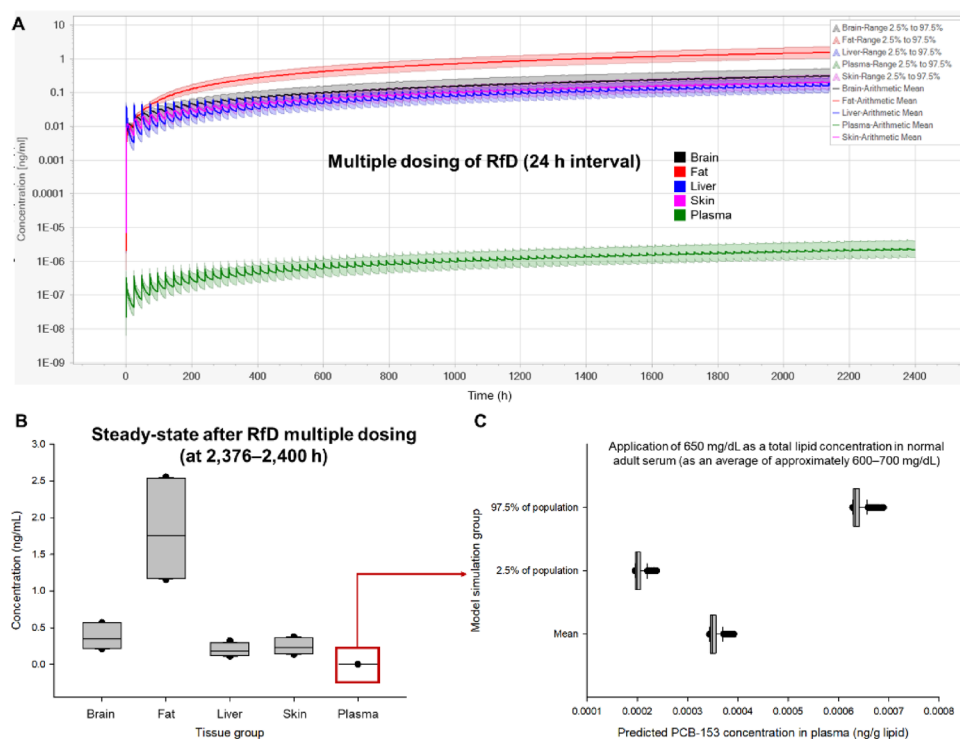


Fig. 7 Results of simulation of the population ($n=100$, comprised of 50% male and 50% female) pharmacokinetic (PK) profiles among target tissues following multiple oral administrations of reference dose (RfD) of PCB-153 using the human physiologically based pharmacokinetic (PBPK) model of PCB-153 established in this study. The solid lines in the graph represent the group mean values predicted by the model, while the shaded regions represent the values corresponding to 2.5–97.5% of the groups, respectively. **A** PK profile results based on model simulation; **B** Comparison of concentration distributions in each target tissue at steady state; **C** Predicted distribution of steady-state plasma concentrations



the general public (McLachlan et al. 2018; ME 2024; Udehman et al. 2018). Thus, the simulation results demonstrate that exposure at the proposed RfD level exhibits consistent distribution characteristics across major tissues within the

human body, providing a scientific basis for predicting body accumulation characteristics, particularly the potential for accumulation in adipose tissue. This can serve as basic data for future exposure assessments and health risk predictions.

Table 4 K_p values of PCB-153 to target tissues at steady state following multiple exposures (24 h dosing interval of RfD) using the human physiologically based pharmacokinetic (PBPK) model of PCB-153 established in this study

Tissue	K_p mean	K_p 2.5% of population	K_p 97.5% of population
Brain	976.96	1054.57	887.95
Fat	4961.45	5725.51	3979.19
Liver	522.61	567.50	472.46
Skin	639.89	690.60	581.51

RfD Reference dose, K_p Tissue distribution coefficient (calculated as the ratio between the concentration value in each tissue and the concentration value in plasma)

Interpretation of tissue-specific toxicokinetic implications

As a result of confirming the distribution and accumulation characteristics of PCB-153 in each tissue based on the steady state standard obtained through multiple repeated exposure simulations (Table 4), the K_p in adipose tissue was the highest at an average of 4961.45, followed by the brain (976.96), skin (639.89), liver (522.61), and plasma (1.0). This suggests that PCB-153, which has high solubility characteristics, may be preferentially distributed to adipose tissue and accumulate in the body for a long time. In particular, accumulation in inactive tissues such as adipose tissue has toxicodynamic implications in that it delays the onset of toxic effects while increasing the possibility of long-term retention in the body.

Table 5 also presents the key PK parameter values under repeated exposure conditions based on the RfD, and these values reflect the within-population variability, including the 2.5–97.5% confidence intervals. In the case of adipose tissue (in the continuous exposure scenario at 24-h intervals), it was impossible to derive parameters such as AUC_{inf} , clearance, MRT, $T_{1/2}$, and V_d due to the continuous PCB-153 tissue accumulation profile characteristics. However, the steady-state C_{trough} was the highest among the target tissues, with an average of 1.76 ng/mL. These results once again demonstrate that adipose tissue is the main accumulation site of PCB-153, supporting the importance of adipose tissue in the in vivo PKs of highly soluble substances. In plasma, the average AUC_{inf} was 0.34 pg·h/mL, showing a relatively low accumulation pattern compared to target tissues such as brain, fat, liver, and skin. In terms of $T_{1/2}$, the estimated values in all target tissues except fat were very long, averaging 8,400–10,500 h, and the MRT was also long, averaging 12,000–15,000 h. The estimated clearance in brain, liver, and skin tissues was very low, averaging 0.003–0.005 mL/min/kg. Overall, these strongly suggested the long-term persistence and low clearance of PCB-153 in the body, and were consistent with the qualitative characterization information known from previous reports (Grandjean et al. 2008; Lindell 2012).

Additionally, this study also presents a schematic diagram of the risk assessment approach using the established human PBPK model, as shown in Fig. 8. This schematic

Table 5 Estimated pharmacokinetic (PK) parameter values from model simulations based on the previously reported RfD of PCB-153 in the human physiologically based pharmacokinetic (PBPK) model of PCB-153 established in this study

Parameter	Unit	Brain	Fat	Liver	Plasma	Skin
AUC_{inf}	ng·h/mL	40.72 [17.14–120.66]	NA	26.53 [10.53–78.86]	3.36E ⁻⁴ [1.20E ⁻⁴ –1.15E ⁻³]	30.32 [12.15–91.31]
AUC_{all}	ng·h/mL	1.44 [0.96–2.23]	5.23 [3.67–7.22]	0.84 [0.54–1.26]	1.00E ⁻⁵ [6.14E ⁻⁶ –1.63E ⁻⁵]	0.96 [0.66–1.50]
C_{max}	ng/mL	1.17E ⁻² [1.04E ⁻² –1.28E ⁻²]	1.77E ⁻² [1.18E ⁻² –2.55E ⁻²]	4.04E ⁻² [2.81E ⁻² –5.02E ⁻²]	3.53E ⁻⁷ [2.41E ⁻⁷ –5.20E ⁻⁷]	1.21E ⁻² [1.12E ⁻² –1.31E ⁻²]
Total body clearance/F	mL/min/kg	3.17E ⁻³ [8.10E ⁻⁴ –5.71E ⁻³]	NA	4.98E ⁻³ [1.24E ⁻³ –9.27E ⁻³]	414.23 [85.04–812.16]	4.35E ⁻³ [1.07E ⁻³ –8.03E ⁻³]
MRT	h	1.20E ⁴ [4.17E ³ –3.00E ⁴]	NA	1.50E ⁴ [4.81E ³ –3.85E ⁴]	1.49E ⁴ [4.77E ³ –3.84E ⁴]	1.38E ⁴ [4.52E ³ –3.54E ⁴]
T_{max}	h	9.43 [6.50–12.38]	NA	1.44 [1.15–1.93]	0.71 [0.40–1.25]	5.12 [3.62–6.75]
$T_{1/2}$	h	8.40E ³ [2.95E ³ –2.08E ⁴]	NA	1.05E ⁴ [3.40E ³ –2.68E ⁴]	1.04E ⁴ [3.37E ³ –2.67E ⁴]	9.66E ³ [3.20E ³ –2.46E ⁴]
V_d (plasma)/F	mL/kg	NA	NA	NA	2.82E ⁸ [1.44E ⁸ –4.61E ⁸]	NA
C_{trough}	ng/mL	0.34 [0.21–0.57]	1.76 [1.17–2.55]	0.18 [0.11–0.30]	2.23E ⁻⁶ [1.27E ⁻⁶ –4.08E ⁻⁶]	0.22 [0.14–0.37]

Parameter values are presented as ‘Mean [range 2.5–97.5% of population]’

RfD Reference dose, NA Not applicable, AUC_{inf} Area under the curve from 0 to infinity after administration, AUC_{all} Area under the curve from 0 to the observation time after administration, C_{max} Maximum concentration, F Bioavailability, MRT Mean residence time, T_{max} Time to reach the maximum concentration, $T_{1/2}$ Half-life, V_d Volume of distribution, C_{trough} Lowest concentration at steady state after multiple doses

^a This is confirmed to be derived based on mouse PK experimental data

^b This is confirmed by estimates obtained through modeling

^c Rough estimate

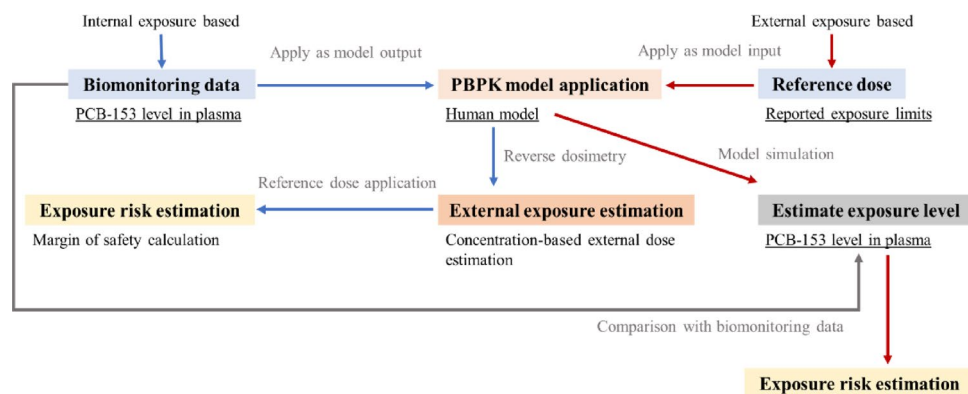


Fig. 8 Schematic diagram of an approach to population exposure risk assessment utilizing human biomonitoring data for PCB-153 using an established human physiologically based pharmacokinetic (PBPK) model for PCB-153. The blue arrow paths in the schematic represent the risk estimation process through the transition flow from internal exposure-based starting values to external exposure estimates, while

the red arrow paths represent the exposure risk estimation process through the transition flow from external exposure-based starting values to internal exposure estimates. The gray arrow paths in the schematic represent the exposure risk estimation process through the direct comparison between monitored internal exposure values and model-based estimated internal exposures

diagram illustrates how the PBPK model can be used for risk assessment by linking human biomonitoring data based on internal exposure. The first approach is to set the existing RfD value as the model input, predict the corresponding concentration level in the target tissue (e.g., blood lipid standardized concentration), and compare this with the actual biomonitoring value to evaluate the potential risk of internal exposure based on the RfD standard. The second approach, conversely, applies biomonitoring data as output from the PBPK model, derives the external exposure required to induce the corresponding internal exposure through reverse dosimetry, and then estimates the safety of exposure by comparing this exposure to the RfD reference value. This structure is designed to significantly contribute to quantitative risk assessment, particularly in calculating external exposures that reflect population-specific characteristics and estimating margins of safety. Consequently, the tissue-specific K_p values and PK parameters derived from this study can be used as basis for predicting the toxicity sensitivity and biological response of each tissue beyond the simple degree of material accumulation, and the established human PBPK model has shown high utility as a quantitative support system for future biomonitoring-based risk assessment.

Discussion

This study aimed to establish a PBPK model based on animal experimental results for PCB-153, a highly chlorinated polychlorinated biphenyl, and to quantitatively predict toxicokinetic exposure at the level of target tissues in the human body through interspecies extrapolation between animals and humans. In particular, this model was designed to predict body accumulation trends in several major target

tissues, such as fat, liver, brain, skin, and plasma, beyond the general single environment or limited tissue levels of existing hazardous substances. It is differentiated from existing universal models in that it presents results that reflect inter-individual (including gender) variations within a population through population-based simulation. The overall model architecture, including the compartmental structure and flow paths, was implemented using the default whole-body PBPK template of PK-Sim®, comprising major organs (e.g., adipose tissue, liver, brain, skin, etc.) interconnected by physiological blood and lymph flows.

Simulation results showed that, under a single oral exposure of 20 mg/kg, adipose tissue exhibited an exceptionally high distribution (mean $K_p = 795,725$ at 336 h), followed by brain (151,257), skin (98,656), liver (79,633), and plasma (1.0) (Fig. 5). This adipose tissue-centered accumulation pattern is interpreted as being due to the physicochemical properties of PCB-153, such as its high solubility (Log P of approximately 7.0; Table 3), which emphasizes its considerably long $T_{1/2}$ and limited biological clearance (Table 5). This is consistent with previous animal studies that have reported similar results, which indicate that PCB-153 can persist in fat for years (Klocke and Lein 2020; Lee et al. 2007, 2002). The results of this study represent one of the first examples to numerically quantify this effect at the human level.

Meanwhile, in the simulation results for repeated administration conditions based on the RfD, which is the human health risk assessment standard, the AUC_{inf} of brain tissue at steady state was derived as an average of 40.72 ng·h/mL, followed by skin (30.32 ng·h/mL), liver (26.53 ng·h/mL), and plasma (0.34 pg·h/mL) (Table 5). Although the AUC_{inf} of adipose tissue could not be estimated to the infinite range due to the profile characteristics, when calculated based

only on AUC_{all} (AUC to observed time not infinity), it was the highest among the target tissues at $5.23 \text{ ng}\cdot\text{h/mL}$, which was approximately four times that of brain tissue. These results suggest that there is a sufficient possibility of long-term accumulation centered on fat even under the RfD standard, and support the need for a conservative interpretation in risk assessment, as it is difficult to rule out the possibility of long-term toxicity due to tissue accumulation even at low exposure doses.

Furthermore, this study schematically illustrates the potential expansion of this PBPK model into a quantitative risk assessment tool (Fig. 8). In addition to the existing RfD-based top-down approach, a bottom-up (reverse dosimetry) approach based on biomonitoring data can be used to predict external exposure equivalent to internal exposure, thereby quantifying the individual margin of safety. This demonstrates the strength of the PBPK model in supplementing the scientific validity of toxicity risk assessments in situations where existing qualitative or quantitative data are limited.

However, this study also has several limitations. First, although the metabolic pathway of PCB-153 mostly involves Phase I metabolism (CYP450 enzymes) and Phase II conjugation metabolism in the liver, information on the precise metabolic rate or concentration of metabolites produced in humans is lacking. Therefore, the metabolic rate in this model was set to “minimal metabolism” based on previously reported material characteristic information (Grandjean et al. 2008; Lindell 2012). This limits its extension to the assessment of the degree of polymorphism in specific metabolite production and toxicity phenotypes. Second, although prediction of lipid distribution is important for the PK properties of high-soluble substances, there is a practical limitation in that there is a significant lack of human biomonitoring data that can be compared with the values measured in adipose tissue in actual clinical settings. To address these limitations, future research emphasizes the need for personalized PBPK modeling, which incorporates quantitative analysis of specific metabolites or reflects individual physiological indicators such as body fat percentage. Furthermore, comparative evaluations with other PCB congeners and integration with mixture toxicity models will be the next steps to better reflect real-world environmental exposure scenarios. In conclusion, this study successfully extrapolated a human PBPK model from animal experimental data, providing quantitative exposure predictions at the target tissue level and risk assessment information that takes into account population variation, making significant academic and policy contributions. In particular, the toxicokinetic accumulation characteristics centered on adipose tissue can serve as important supporting data for future studies on chronic health effects and the establishment of protective limits following PCB-153 exposure.

Furthermore, the RfD, the external exposure criterion applied in the single and repeated exposure simulations conducted in this study, was based on values presented in recent literature (Ermler and Kortenkamp 2022), and the RfD was calculated by setting sperm function impairment as the main toxicity indicator in animal experiments. In this study, the predicted PCB-153 concentrations in major target tissues (fat, brain, liver, skin, etc.) in humans as well as in plasma were significantly lower (Fig. 7) than the actual internal exposure levels reported in previously reported human blood biomonitoring data (Crinnion 2010; McLachlan et al. 2018; ME 2024; Undeman et al. 2018). This implies that the applied RfD itself may have been estimated conservatively or excessively low compared to the actual toxicity level in humans. In particular, considering the characteristics of highly soluble and bioaccumulative substances such as PCB-153, there is concern that the RfD derived solely from extrapolated animal toxicity data may not adequately reflect the actual human exposure situation. Therefore, future studies will require resetting the RfD based on more precise toxicodynamic and pharmacodynamic scientific evidence, and based on this, it is expected that a more reliable risk assessment of PCB-153 will be possible through linkage with human biomonitoring data.

Conclusion

This study established a human PBPK model for the environmental persistent pollutant PCB-153 and quantitatively predicted tissue-level accumulation patterns and exposure based on this model, thereby presenting a quantitative tool that can overcome the limitations of existing risk assessment approaches. In particular, adipose tissue exhibited a distinct accumulation characteristic, with an average K_p of over 79,633 among target tissues after a single oral dose of 20 mg/kg . Even under repeated exposure conditions (RfD), the average K_p in adipose tissue was the highest at steady state, at 4,961, numerically confirming the possibility of long-term retention, particularly in high-adipose tissue. Furthermore, this model demonstrated the feasibility of realistic exposure predictions considering diverse physiological characteristics within the human body through Monte Carlo simulations that reflected interindividual variation within a population. These results not only deepen our understanding of the toxicokinetics of PCB-153 but also provide a feasible framework for top-down and bottom-up quantitative risk assessment strategies based on internal exposure, demonstrating significant academic and practical value. This study can be used as scientific basis for the future assessment of human impact and establishment of safety standards for highly soluble environmental pollutants, including

PCB-153, and is significant as a case that demonstrates the practical applicability of PBPK-based quantitative toxicity assessment.

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Data availability All data and related materials are accessible in this manuscript.

Declarations

Conflict of interest The authors have no conflicts of interest relevant to this study to disclose.

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